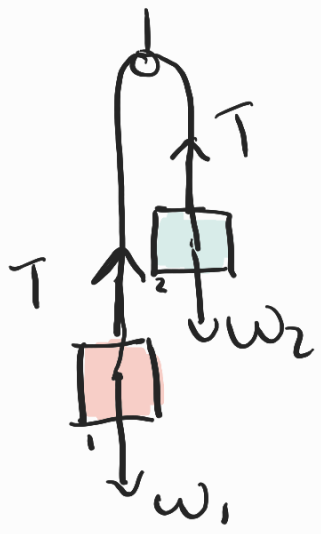
①



$m_1 = 3kg$
$m_2 = 5kg$

a) $\Sigma F_{y_1} = m_1 a = T - w_1$

$\Rightarrow m_1 a = T - m_1 g \Rightarrow T = m_1 a + m_1 g$ ①

$\Sigma F_{y_2} = m_2 a = w_2 - T$

$\Rightarrow m_2 a = m_2 g - T$ ②

① → ② : $m_2 a = m_2 g - m_1 a - m_1 g$

$\Rightarrow 5a = 50 - 3a - 30$

$\Rightarrow 8a = 20 \Rightarrow a = 2.5 m/s^2$ //

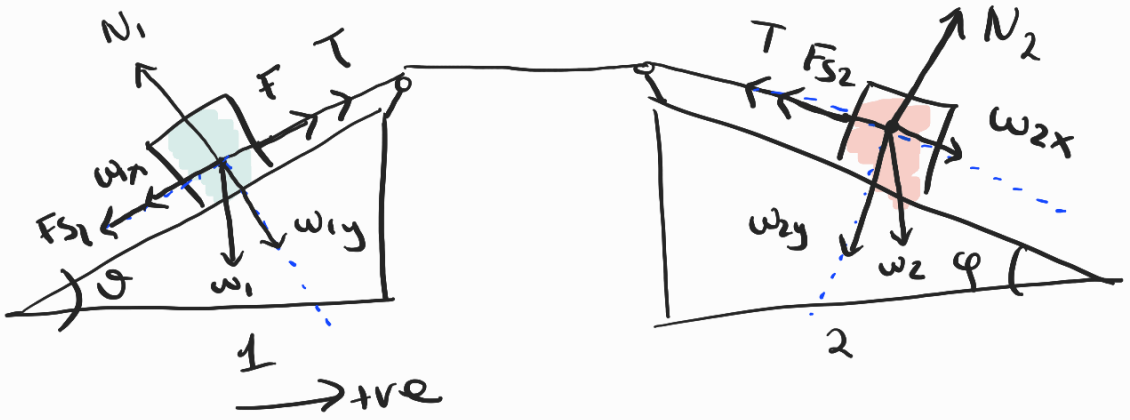
(b) $T = m_1 a + m_1 g = 3(2.5) + 3 \cdot (10)$

$\Rightarrow T \cong 37N$ //

(c) $x = x_0 + u_0 t + \frac{1}{2} a t^2$

$\Rightarrow x = 0 + 0 + \frac{1}{2}(2.5)(2)^2 \cong 5m$ //

②



$\theta = 30^\circ$ $\quad F = 10N$ $\quad W_{1x} = m_1 g \sin\theta$

$\varphi = 45^\circ$ $\quad W_{1y} = m_1 g \cos\theta$

$m_1 = 3kg$ $\quad W_{2x} = m_2 g \sin\varphi$

$\mu_1 = 0.2$ $\quad m_2 = 4kg$ $\quad W_{2y} = m_2 g \cos\varphi$

$\mu_2 = 0.3$ $\quad g = 10\ m/s^2$

(a) Σ ωka m_1 :

$$\Sigma F_{1x} = m_1 a = F + T - W_x - F_{s1}$$

$$F + T - m_1 g \sin\theta - \mu_1 N_1 = m_1 a \quad ①$$

$$\Sigma F_{1y} = 0 \Rightarrow N_1 = W_{1y} = m_1 g \cos\theta \quad ②$$

② → ①: $F + T - m_1 g \sin\theta - \mu_1 m_1 g \cos\theta = m_1 a$

$\Rightarrow 10 + T - (3)(10)\sin 30 - (0.2)(10)(3)\cos 30 = 3a$

$10 + T - 15 - 5.2 = 3a$

$$\Rightarrow \boxed{T = 3a + 10.2} \quad ③$$

Σ ωfa m_2

$$\Sigma F_{2x} = m_2 a = w_{2x} - f_{s2} - T$$

$$\Rightarrow m_2 g \sin\theta - \mu_{s2} N_2 - T = m_2 a \quad ④$$

$$\Sigma F_{2y} = 0 \Rightarrow N_2 = w_y = m_2 g \cos\varphi \quad ⑤$$

⑤ → ④ : $m_2 g \sin\varphi - \mu_{s2} m_2 g \cos\varphi - T = m_2 a$

$\Rightarrow (4)(10)\sin 45 - (0.3)(4)(10)\cos 45 - T = 4a$

$28.3 - 8.5 - T = 4a$

$$\Rightarrow \boxed{T = 19.8 - 4a} \quad ⑥$$

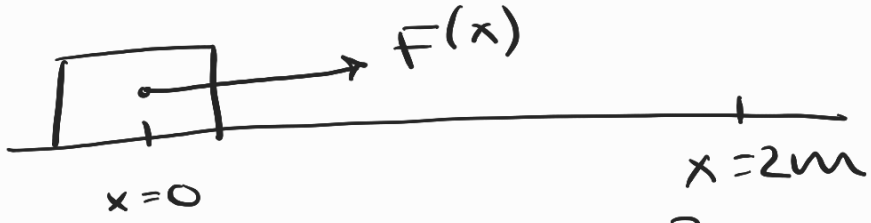
③ = ⑥ : $3a + 10.2 = 19.8 - 4a$

$\Rightarrow 7a = 9.6$

$\Rightarrow a = 1.37 \text{ m/s}^2$

(b) ③ : $T = 3(1.37) + 10.2 = 14.31 \text{ N}$

③ $F(x) = 3x^2$



(a) $W = \int_0^2 F(x)dx = \left[\frac{3x^3}{3}\right]_0^2 = 2^3 - 0^3 = 8\,J$ //

(b) $m = 2\,kg$, $\Delta E_k = ?$

$W = \Delta E_k$

υποθέτουμε ότι το σώμα ξεκινάει από ηρεμία

$\Rightarrow \Delta E_k = E_{kf} - \cancel{E_{ki}}^{\,0} = 8\,J$

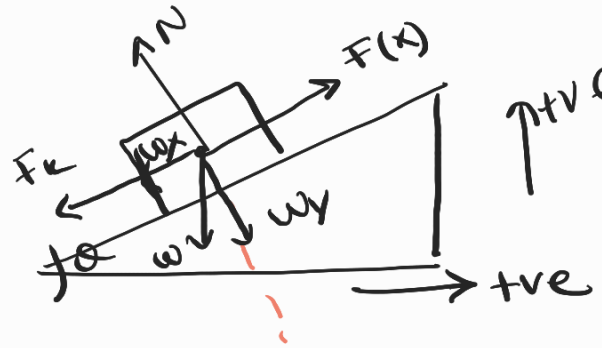
(c) $v = ?$

$v_0 = 0\,m/s$

$E_k = \frac{1}{2}mv^2 \Rightarrow 8 = \frac{1}{2}(2)v^2$

$\Rightarrow v = \sqrt{8} = 2\sqrt{2} = 2.83\,m/s$ //

④



$\theta = 30°$

$F(x) = 100 - 10x$

$m = 10\ kg$

$\mu_k = 0.3$

(a) $W_F = ?$ $\quad x_0 = 0\ m$, $\quad x = 4\ m$

$$W_F = \int_{x_0}^{x} F(x)\,dx = \int_0^4 100 - 10x$$

$$= \left[100x - \frac{10x^2}{2}\right]_0^4$$

$$= 100(4) - \frac{10(4)^2}{2} - 0$$

$$\Rightarrow W_F = 400 - 5(16) = 320\ J$$

b) $f_k = \mu_k \cdot N$

$\Sigma F_y = 0 \Rightarrow N = W_y = W\cos\theta$

$\Rightarrow f_k = \mu_k \cdot mg\cos\theta \simeq 25.5\ N$

$W_{f_k} = -f_k \cdot d = -(25.5)(4) = -102\ J$

(c) $W_w = ?$

Μόνο W_x παράγει έργο γιατί W_y είναι κάθετη στην κίνηση

$$W_x = mg\sin\theta$$

$$\Rightarrow W_{wx} = -W_x \cdot d = -mg\sin\theta\,(4)$$

$$\simeq -10(10)\sin 30\,(4)$$

$$= -400\,(0.5) = -200\,J$$

(d) $W_{TOT} = \Delta E_k$

$$W_{TOT} = W_F + W_w + W_{Fk}$$

$$= 320 - 200 - 102$$

$$= 18\ J$$

$$\Rightarrow 18 = \frac{1}{2}mv^2 - \frac{1}{2}mv_0^2$$ (το v_0 είναι 0)

$$\Rightarrow v^2 = \frac{2 \cdot 18}{m} \Rightarrow v = \sqrt{\frac{2 \cdot 18}{10}}\ m/s$$

$$\Rightarrow v \simeq 1.9\ m/s$$